1. The common name of ethanedioic acid is

(a) Acetic acid **(b) Oxalic acid** (c) Adipic acid (d) Succinic acid

2. Aldehyde which do not respond to Fehling's test is

(a) Formaldehyde (b) Ethanal **(c) Benzaldehyde** (d) Mesityl oxide

3. Salicylaldehyde can be prepared from which of the following reactants?

(a) Phenol and Chloroform **(b) Phenol, Chloroform and aq. NaOH**

(c) Phenol and CCl4 (d) Phenol, CCl4 and NaOH

4. Hydration of acetylene (ethyne) with dil. H_2SO_4 and $HgSO_4$ gives

(a) Acetone (b) 2-phenylethanol **(c) Acetaldehyde** (d) Acetic acid

5. The best reagent useful for separation and purification of aldehyde from ketone is

a) Tollens' reagent **b) sodium hydrogen sulphite**

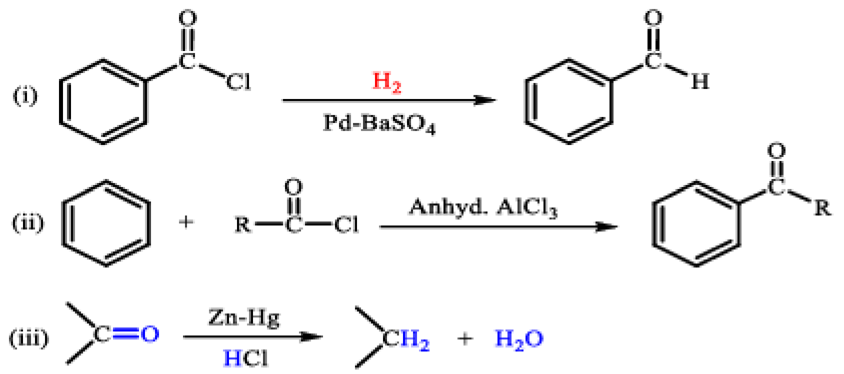
c) sodium sulphate d) 2,4-DNP reagent

6. The reagent which does not react with both, acetone and benzaldehyde.

(a) Sodium hydrogensulphite (b) Phenyl hydrazine

(c) Fehling's solution (d) Grignard reagent

7. Mention the name of the following reactions.

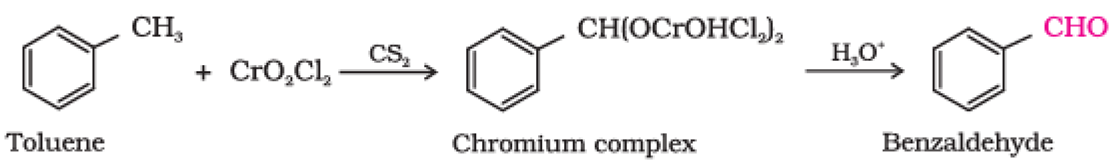


Ans: **i)** Rosenmund's Reduction reaction **ii)** Friedel-Craft's Acylation Reaction

iii) Clemmenson's Reduction

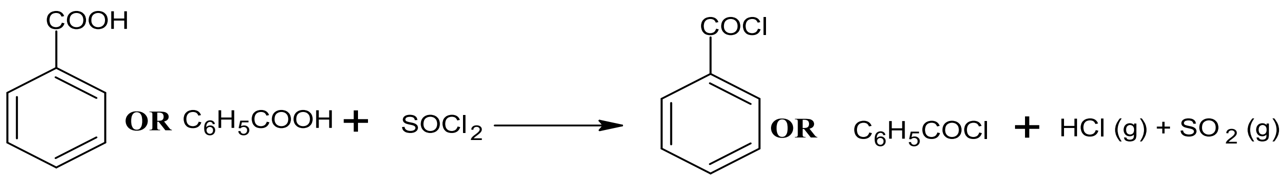
8. Write the reaction for the conversion of toluene to benzaldehyde using chromyl chloride. Name the reaction.

Ans: Etard reaction



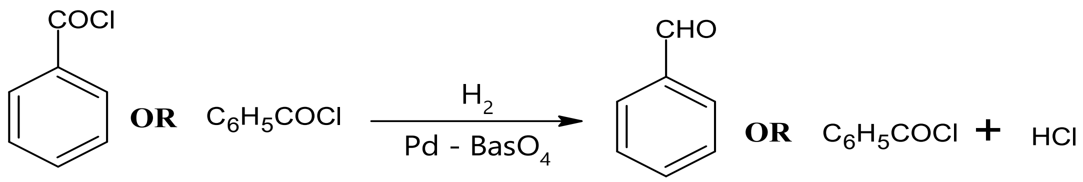
9. An organic compound 'P' ($C_7H_6O_2$) reacts with thionyl chloride gives product 'Q'. 'Q' on the further undergoes hydrogenation in the presence of Pd-$BaSO_4$ forms compound 'R'. Write the equations involved in the chemical reactions.

Ans:



P = Benzoic acid

Q= Benzoyl chloide

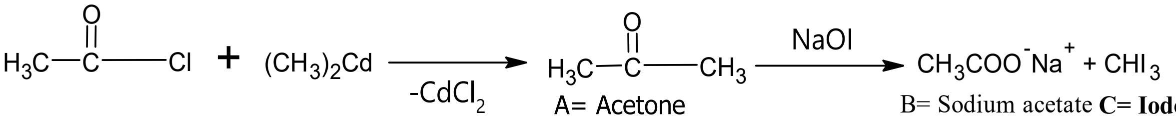


Q= Benzoyl chloide

R = Benzaldehyde

10. Identify the compounds ‘A’, ‘B’ and ‘C’ in the following reaction.

Ans:



A= Acetone

B= Sodium acetate **C= Iodoform OR triiodomethane**

11. Among methanal and ethanal, which one undergoes Cannizaro reaction? Give reason.

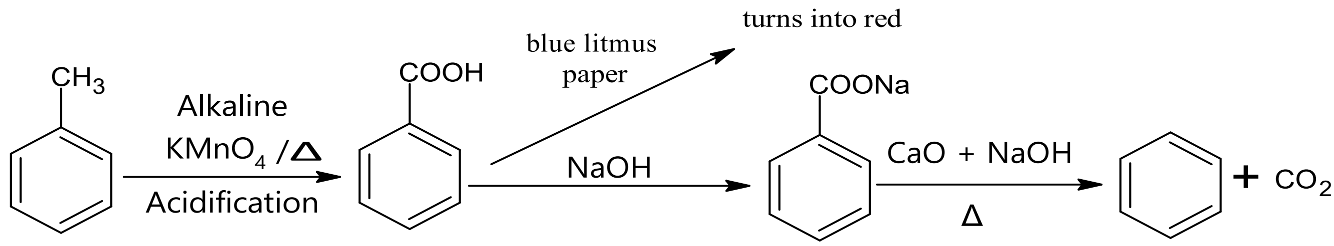
Ans: Methanal undergo Cannizzaro’s reaction due to absence of α (alpha)- hydrogen atom.

12. The compound “A” has molecular formula C_7H_8 is heated with alkaline $KMnO_4$ followed by acidification gives compound “B” of the molecular formula $C_7H_6O_2$. This compound “B” turns blue litmus to red. The sodium salt of compound “B” is heated with reagent “X” gives hydrocarbon “C’ of molecular mass 78 g mol^{-1}.

(a) Write the structure of compounds “A”, “B” and “C”.

(b) Identify “X” and mention its role in the reaction.

Ans: a)



A= C_7H_8
Toluene/
Methyl benzene

B=
$C_7H_8O_2$
B = Benzoic acid

Sodium salt of benzoic acid

C= C_6H_6
Benzene

Benzene

b) CaO + NaOH (Soda lime), act as decarboxylating agent **OR** remove the CO_2 from carboxyl acid

13.

$$CH_3MgBr + A \xrightarrow[\text{ii) } H_3O^+]{\text{i) dry ether}} B \xrightarrow[\text{ii) } H_2O]{\text{i) } Cl_2 / \text{Red P}} C$$

(a) Identify the compound A, B and C in the above reaction.

Ans: A= CO_2 (Dry ice) B = CH_3COOH (Acetic acid) C= $ClCH_2$ COOH (Chloroacetic acid)

(b) Name the reaction for the conversion of compound B to C.

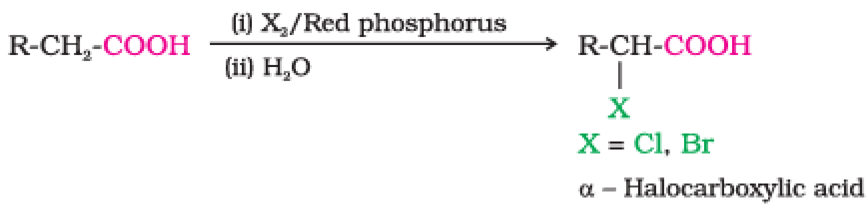
Ans: Hell-Volhard-Zelinsky reaction

(c) Compound 'C' is more acidic than 'B'. Give reason.

Ans: Chlorine **stabilises the carboxylate anion by (-I) inductive effects**

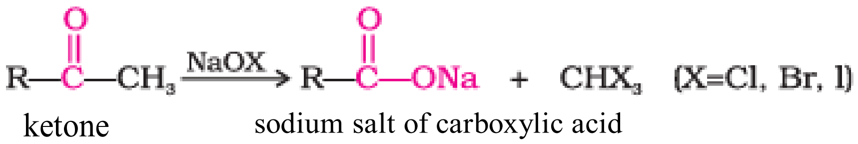
14. Explain Hell-Volhard –Zelinsky reaction with chemical equation.

Ans: Carboxylic acids having an **α-hydrogen** are halogenated at the α-position on treatment with **chlorine or bromine** in the presence of small amount of **red phosphorus** to give **α-halocarboxylic acids.**



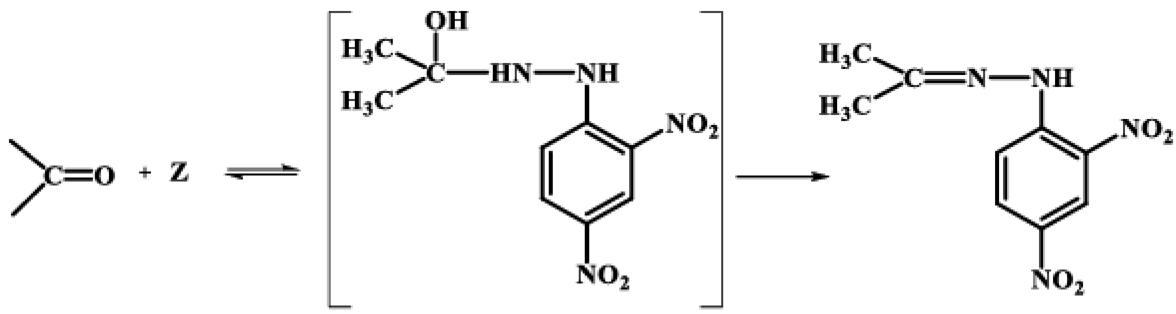
15. Provide the general reaction to prepare Haloform from methyl ketones. Name a reagent which is used to detect the presence of α – methyl group in a carbonyl compound.

Ans:

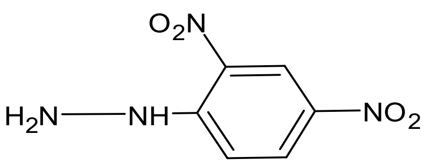


Sodium hypohalite OR sodium hypoiodite

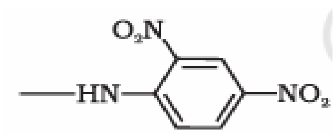
16. In a given chemical reaction, write the structure and name of compound 'Z'.



Ans: Z =



OR



2,4-Dinitrophenyl hydrazine

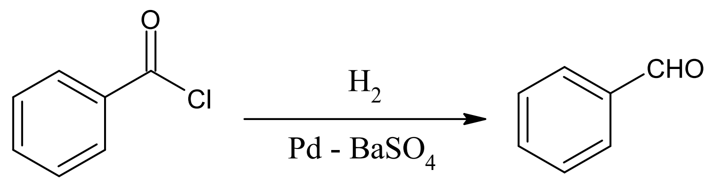
17. Name the gas used to facilitate the nucleophilic addition of ethylene glycol to aldehydes and ketones.

Ans: Dry HCl gas

Few most repeated questions:

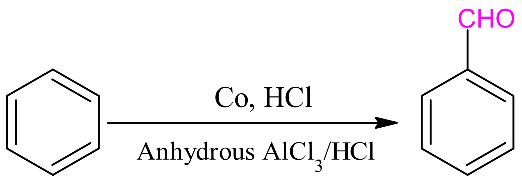
18. Explain Rosenmund reduction of benzoyl chloride. **OR** How is benzaldehyde prepared by Rosenmund reduction? **OR** How is Benzoyl chloride converted into Benzaldehyde? Name the reaction.

Ans:



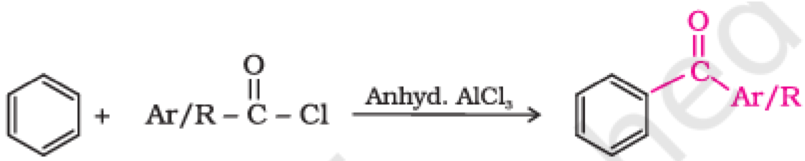
Benzoyl chloride   Benzaldehyde

19. How benzene is converted into benzaldehyde by Gatterman- Koch reaction? Write equation.

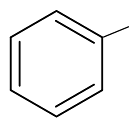


Benzene   Benzaldehyde

20. How does benzene reacts with acetyl – chloride in the presence of anhydrous $AlCl_3$? Give equation.



Benzene   Aryl/Acyl Chloride   Ketone (Acetophenone/Benzophenone)

(R =alky, Ar = 

21. Ketone are generally less reactive towards nucleophilic addition reactions **OR** Aldehydes are more reactive than ketones towards Nucleophilic addition reaction. Give reason

Ans: **Steric** and **Electronic effect** in ketones

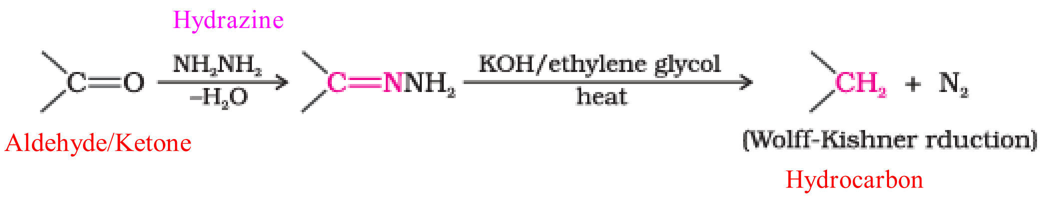
22. Explain Clemmensen reduction reaction.

Ans:

Aldehyde/Ketone   Hydrocarbon

23. Write Wolff-Kishner's reduction for the conversion of Carbonyl group into $–CH_2–$ group.

Ans:

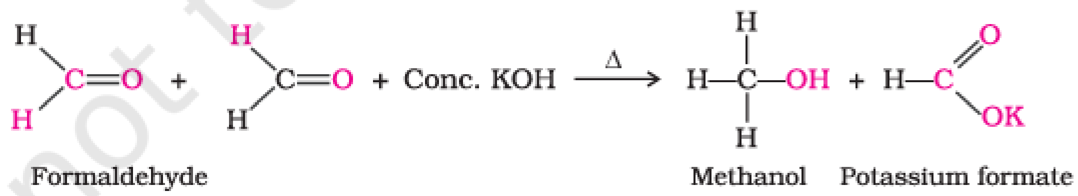


24. Formaldehyde does not undergo aldol condensation. Give reason **OR** Why C_6H_5CHO does not undergo aldol reaction?

Ans: due to absence of α- hydrogen

25. Write equations for : The reaction between formaldehyde and conc. KOH **OR** How does formaldehyde reacts with concentrated alkali on heating? Name this reaction**.**

Ans:

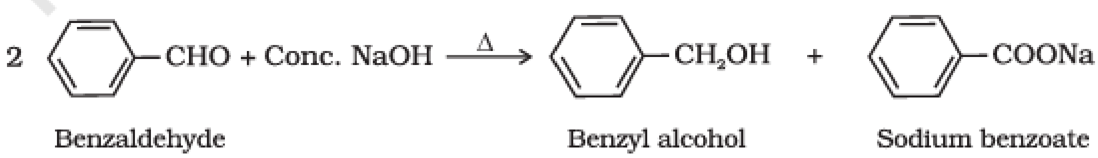


OR

$$2HCHO + Con.KOH \xrightarrow{\Delta} CH_3OH + HCOOK$$

Formaldehyde Methyl alcohol Potassium formate

26. Write the equation for the reaction between benzaldehyde and concentrated NaOH solution. Name the reaction.

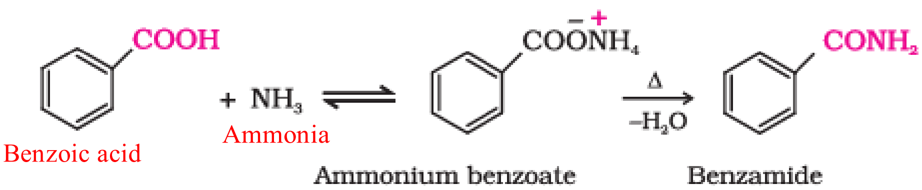


27. What are the effect of electron withdrawing and electron donating groups on the acidity of carboxylic acis?

Ans:

Substituents	**Acidity**	**Reason**
Electronwith drawing group (-NO_2, F, Cl)	Increases	EWG **Stabilizing** the carboxylate ion (-I effect)
Electron donating group (-CH_3 or – CH_3O)	decreases	EDG **destabilizing** carboxylate ion (+I effect)

28. How do you convert benzoic acid to benzamide? Write the reaction



29. Explain esterification reaction with an example.

H_2SO_4 or HCl gas catalyst

$$RCOOH + R'OH \xrightleftharpoons{H^+} RCOOR' + H_2O$$

carboxylic acid Alcohol Ester